

Reference Integration Report

SURF2026 American Option Risk Surfaces

Codex-assisted reference pass

June 17, 2026

Abstract

This report documents the reference integration pass for the completed solver-validation study reports. The pass creates one shared BibTeX file, adds verified citations to the ticket reports, regenerates the report PDFs, and records unresolved source questions in a backlog. It does not modify solver code, numerical results, figures, experiments, datasets, stress maps, neural-network files, or machine-learning labels.

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1 Purpose of the Pass

Tickets 01 through 12 intentionally deferred formal bibliography work while the solver-validation ladder was still being built. This pass adds academic grounding after the validation evidence is complete. Its purpose is narrow:

- cite verified sources for the main theoretical and methodological claims;
- keep technical results and conclusions unchanged;
- avoid inventing metadata or fake DOI records;
- record unresolved source questions for later human review.

2 Source Verification Policy

References were added only when bibliographic metadata could be verified through Crossref/DOI records or stable publisher-style metadata. If metadata was incomplete, conflicting, or merely a reading lead in the project documents, the item was placed in `reference_backlog.md` instead of being guessed into the bibliography.

This policy matters because the solver reports are meant to be study documents for beginner researchers. A small verified bibliography is more useful than a long bibliography with uncertain details.

3 Shared Bibliography File

The shared bibliography is:

`reports/03_solver/references.bib`

The ticket reports use `natbib` with author-year citations and a shared BibTeX database:

```
\usepackage[round,authoryear]{natbib}
\bibliographystyle{plainnat}
\bibliography{reports/03_solver/references}
```

Citation keys use short lowercase identifiers such as `black_scholes_1973`, `crank_nicolson_1947`, and `cottle_pang_stone_2009`.

4 References Added

Citation key	Reference role	Verification note
<code>black_scholes_1973</code>	European option pricing and Black-Scholes model.	DOI metadata verified.
<code>merton_1973</code>	Rational option pricing, dividends, and American-call context.	DOI metadata verified.

Citation key	Reference role	Verification note
crank_nicolson_1947	Crank-Nicolson time stepping.	DOI metadata verified.
brennan_schwartz_1977	American put finite-difference valuation background.	JSTOR DOI metadata verified; alternate record noted in backlog.
rannacher_1984	Startup damping for irregular initial data.	DOI metadata verified.
wilmott_howison_dewynne_1995	Option-PDE, finite-difference, boundary, and Greek background.	DOI metadata verified.
cottle_pang_stone_2009	Linear complementarity problem methodology.	SIAM electronic record verified.
longstaff_schwartz_2001	American option simulation as contrast method.	DOI metadata verified.
haugh_kogan_2004	American option duality bounds as background.	DOI metadata verified.
raissi_perdikaris_karniadakis_2019	PINN future-work context.	DOI metadata verified.
sirignano_spiliopoulos_2018	Deep Galerkin Method future-work context.	DOI metadata verified.
becker_cheridito_jentzen_2020	Deep learning for American-style options.	DOI metadata verified.
gatta_etal_2023	PINN American option future-work context.	DOI metadata verified.
nwankwo_etal_2024	Neural free-boundary American option future-work context.	DOI metadata verified.

5 Reference-to-Report Mapping

Report	Main citation topics	Representative keys
Ticket 01	Black-Scholes and Merton closed-form pricing.	black_scholes_1973, merton_1973.
Ticket 02	Black-Scholes operator and finite-difference setup.	black_scholes_1973, brennan_schwartz_1977, wilmott_howison_dewynne_1995.
Ticket 03	European CN validation.	crank_nicolson_1947, black_scholes_1973.
Ticket 04	American obstacle, LCP, PSOR core.	brennan_schwartz_1977, cottle_pang_stone_2009.
Ticket 05	American put validation.	brennan_schwartz_1977, cottle_pang_stone_2009.
Ticket 06	No-dividend American call control.	black_scholes_1973, merton_1973.
Ticket 07	Dividend-paying American call behavior.	black_scholes_1973, merton_1973, cottle_pang_stone_2009.

Report	Main citation topics	Representative keys
Ticket 08	Obstacle and complementarity diagnostics.	cottle_pang_stone_2009.
Ticket 09	Continuation premium and boundary extraction.	brennan_schwartz_1977, wilmott_howison_dewynne_1995.
Ticket 10	Delta/Gamma diagnostics.	wilmott_howison_dewynne_1995, rannacher_1984.
Ticket 10A	Rannacher comparison gate.	crank_nicolson_1947, rannacher_1984.
Ticket 11	Grid and domain sensitivity.	crank_nicolson_1947, wilmott_howison_dewynne_1995.
Ticket 12	Solver-validation synthesis and future-work context.	Classical solver, LCP, Rannacher, and selected scientific-ML references.

6 Unresolved Backlog Summary

The backlog file is:

`reports/03_solver/reference_backlog.md`

The main unresolved items are:

- Brennan-Schwartz has a verified JSTOR DOI record and a conflicting Wiley metadata record; the JSTOR record was used and the conflict was recorded.
- Tavella and Randall was considered but not added because the Crossref search surfaced a review rather than a verified book record.
- Some future-work neural and process-informed sources from the literature map remain reading leads rather than necessary citations for the solver-validation reports.
- Advisor review may request denser citation coverage in a future paper-style literature review.

7 Compilation Status

The updated ticket reports and this integration report are intended to compile with `latexmk-xelatex-bibtex`. The implementation pass checks for unresolved citation warnings, confirms that each regenerated PDF exists, and uses PDF text extraction to confirm that reference sections appear in the compiled reports.

Report group	Expected status	Check
Ticket 01 through Ticket 12 PDFs	Compiled	XeLaTeX/BibTeX via <code>latexmk</code> .
Reference integration report PDF	Compiled	XeLaTeX/BibTeX via <code>latexmk</code> .
Citation resolution	No unresolved citations	Log search after compilation.

Report group	Expected status	Check
Bibliography visibility	Present	PDF text extraction and spot checks.

8 Limitations

This pass is not a final paper literature review. It adds the minimum verified references needed to ground the solver-validation reports. It does not decide the final paper's citation style, does not expand every theoretical aside into a literature survey, and does not convert diagnostic boundary or Greek outputs into production claims.

9 Recommended Next Step

The recommended next step is human review: confirm that the citation placement is sufficient for the course or supervisor's expectations, resolve the backlog items that matter for the final paper, and only then decide whether to begin downstream stress-map, dataset, or surrogate-model work.

References

- Sebastian Becker, Patrick Cheridito, and Arnulf Jentzen. Pricing and hedging american-style options with deep learning. *Journal of Risk and Financial Management*, 13(7):158, 2020. doi: 10.3390/jrfm13070158.
- Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.
- Michael J. Brennan and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449, 1977. doi: 10.2307/2326779.
- Richard W. Cottle, Jong-Shi Pang, and Richard E. Stone. *The Linear Complementarity Problem*. Society for Industrial and Applied Mathematics, 2009. ISBN 9780898716863. doi: 10.1137/1.9780898719000.
- J. Crank and P. Nicolson. A practical method for numerical evaluation of solutions of partial differential equations of the heat-conduction type. *Mathematical Proceedings of the Cambridge Philosophical Society*, 43(1):50–67, 1947. doi: 10.1017/S0305004100023197.
- Federico Gatta, Vincenzo Schiano Di Cola, Fabio Giampaolo, Francesco Piccialli, and Salvatore Cuomo. Meshless methods for american option pricing through physics-informed neural networks. *Engineering Analysis with Boundary Elements*, 151:68–82, 2023. doi: 10.1016/j.enganabound.2023.02.040.
- Martin B. Haugh and Leonid Kogan. Pricing american options: A duality approach. *Operations Research*, 52(2):258–270, 2004. doi: 10.1287/opre.1030.0070.
- Francis A. Longstaff and Eduardo S. Schwartz. Valuing american options by simulation: A simple least-squares approach. *Review of Financial Studies*, 14(1):113–147, 2001. doi: 10.1093/rfs/14.1.113.
- Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141, 1973. doi: 10.2307/3003143.

- Chinonso Nwankwo, Nneka Umeorah, Tony Ware, and Weizhong Dai. Deep learning and american options via free boundary framework. *Computational Economics*, 64(2):979–1022, 2024. doi: 10.1007/s10614-023-10459-3.
- M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019. doi: 10.1016/j.jcp.2018.10.045.
- Rolf Rannacher. Finite element solution of diffusion problems with irregular data. *Numerische Mathematik*, 43(2):309–327, 1984. doi: 10.1007/BF01390130.
- Justin Sirignano and Konstantinos Spiliopoulos. Dgm: A deep learning algorithm for solving partial differential equations. *Journal of Computational Physics*, 375:1339–1364, 2018. doi: 10.1016/j.jcp.2018.08.029.
- Paul Wilmott, Sam Howison, and Jeff Dewynne. *The Mathematics of Financial Derivatives*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511812545.